MATHS MANTHAN
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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 03
Chapter 1: Real Numbers

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

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Q1. If p is a prime number, then the LCM of p and p^2 is: [1]

(a) p (b) p^2 (c) p^3 (d) 1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) p^2

Since $p = p^1$ divides p^2, we apply the LCM highest-power rule:

Formula Used: $\text{LCM}(p^1, p^2) = p^{\max(1,2)}$

Since $\max(1, 2) = 2$, we get $\text{LCM}(p, p^2) = \boxed{p^2}$. [1M]

Q2. If $\text{HCF}(a, b) = 12$ and $a \times b = 1800$, then $\text{LCM}(a, b)$ is: [1]

(a) 3600 (b) 150 (c) 900 (d) 600

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 150

Given $\text{HCF}(a, b) = 12$ and $a \times b = 1800$. We know:

Formula Used: $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

$$\text{LCM}(a, b) = \frac{1800}{12} = \boxed{150}.$$ [1M]

Q3. If n is any natural number, then $6^n - 5^n$ always ends with the digit: [1]

(a) 1 (b) 3 (c) 5 (d) 7

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 1

We know unit-digit cyclicity for powers ending in 6 and 5:

Formula Used: $\forall\, n \in \mathbb{N}$, 6^n ends with 6 and 5^n ends with 5

Hence the unit digit of $6^n - 5^n$ is $6 - 5 = \boxed{1}$ for every natural number n (e.g., $n = 1 : 6 - 5 = 1$; $n = 2 : 36 - 25 = 11$). [1M]

Q4. If n is a natural number, then $(9^{2n} - 4^{2n})$ is always divisible by: [1]

(a) 5 only (b) 13 only (c) both 5 and 13 (d) neither 5 nor 13

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (c) both 5 and 13

Rewriting $9^{2n} - 4^{2n} = (9^2)^n - (4^2)^n = 81^n - 16^n$. We know the standard factorisation identity:

Formula Used: $a^n - b^n$ is always divisible by $(a - b)$ $\forall\, n \in \mathbb{N}$

Hence $81^n - 16^n$ is divisible by $81 - 16 = 65 = 5 \times 13$ (for $n = 1$: $9^2 - 4^2 = (9-4)(9+4) = 5 \times 13 = 65$). Thus it is always divisible by $\boxed{\text{both 5 and 13}}$. [1M]

Q5. Assertion (A): The number 6^n cannot end with the digit 0 for any natural number $n \in \mathbb{N}$.
Reason (R): The prime factorisation of 6 has only 2 and 3 as prime factors, and does not contain the prime factor 5. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

We know the Fundamental Theorem of Arithmetic condition for ending with 0:

Formula Used: $10 = 2 \times 5$ must divide a number ending in 0

Because $6 = 2 \times 3$, $6^n = 2^n \times 3^n$ lacks prime factor 5. Thus both (A) and (R) are true and (R) explains (A). [1M]

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SECTION B: Very Short Answer Type

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Q6. Check whether 4^n can end with the digit 0 for any natural number $n \in \mathbb{N}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

We know the prime-factor criterion for trailing zeros via the Fundamental Theorem of Arithmetic:

Formula Used: If 4^n ends with 0, it must be divisible by $10 = 2 \times 5$

Writing 4^n in prime factorised form for any $n \in \mathbb{N}$:

$$4^n = (2^2)^n = 2^{2n}. \qquad [1M]$$

Only prime 2 appears in the prime factorisation of 4^n; prime 5 is missing. By the uniqueness of FTA, 5 cannot divide 4^n. Therefore, $\boxed{4^n \text{ cannot end with digit 0 for any } n \in \mathbb{N}.}$ [1M]

SECTION C: Short Answer Type

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Q7. Prove that $\sqrt{3} + \sqrt{5}$ is an irrational number. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Suppose, on the contrary, that $r = \sqrt{3} + \sqrt{5}$ is rational ($r \in \mathbb{Q},\ r > 0$). Isolating $\sqrt{5} = r - \sqrt{3}$ and squaring both sides:

$$(\sqrt{5})^2 = (r - \sqrt{3})^2 \implies 5 = r^2 + 3 - 2r\sqrt{3}. \qquad [1M]$$

Rearranging terms to solve for $\sqrt{3}$:

$$2r\sqrt{3} = r^2 + 3 - 5 = r^2 - 2 \implies \sqrt{3} = \frac{r^2 - 2}{2r}.$$

We know rational numbers are closed under basic arithmetic (with non-zero denominator):

Formula Used: $r \in \mathbb{Q} \setminus \{0\} \implies \dfrac{r^2 - 2}{2r} \in \mathbb{Q}$

Hence $\sqrt{3}$ must be rational. [1M]

However, $\sqrt{3}$ is irrational (if $\sqrt{3} = \frac{a}{b}$ with HCF$(a, b) = 1$, then $a^2 = 3b^2 \implies 3 \mid a$ and $3 \mid b$, contradiction). Thus our assumption is false, proving $\boxed{\sqrt{3} + \sqrt{5} \text{ is irrational.}}$ [1M]

Q8. Prove that $\dfrac{2+3\sqrt{5}}{7}$ is an irrational number, given that $\sqrt{5}$ is irrational. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: $\sqrt{5}$ is irrational. Suppose, on the contrary, that $\dfrac{2+3\sqrt{5}}{7}$ is rational, equal to $\frac{a}{b}$ where $a, b \in \mathbb{Z}$ and $b \neq 0$:

$$\frac{2+3\sqrt{5}}{7} = \frac{a}{b} \implies 2+3\sqrt{5} = \frac{7a}{b}. \qquad \text{[1M]}$$

We know the closure property of rational numbers under subtraction and non-zero division:

Formula Used: $x, y \in \mathbb{Q},\ y \neq 0 \implies x - y \in \mathbb{Q}$ and $\dfrac{x}{y} \in \mathbb{Q}$

Isolating $\sqrt{5}$:

$$3\sqrt{5} = \frac{7a}{b} - 2 = \frac{7a-2b}{b} \implies \sqrt{5} = \frac{7a-2b}{3b}. \qquad \text{[1M]}$$

Since $a, b \in \mathbb{Z}$ and $3b \neq 0$, $\frac{7a-2b}{3b}$ is a rational number, contradicting the given fact that $\sqrt{5}$ is irrational.

Hence $\boxed{\dfrac{2+3\sqrt{5}}{7} \text{ is irrational.}}$ **[1M]**

Q9. Prove that $3+\sqrt{2}$ is irrational. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Suppose, on the contrary, that $3+\sqrt{2}$ is rational. Then there exist co-prime integers a, b $(b \neq 0)$ such that:

$$3+\sqrt{2} = \frac{a}{b} \implies \sqrt{2} = \frac{a}{b} - 3 = \frac{a-3b}{b}. \qquad \text{[1M]}$$

We know rational numbers are closed under subtraction and prime square-root irrationality:

Formula Used: $a, b \in \mathbb{Z},\ b \neq 0 \implies \dfrac{a-3b}{b} \in \mathbb{Q}$, while $\sqrt{2} \notin \mathbb{Q}$

Thus $\sqrt{2}$ must be rational. **[1M]**

Yet $\sqrt{2}$ is irrational (if $\sqrt{2} = \frac{u}{v}$ with $\text{HCF}(u, v) = 1$, $u^2 = 2v^2 \implies 2 \mid u$ and $2 \mid v$, contradiction).

Hence our initial supposition fails, so $\boxed{3+\sqrt{2} \text{ is irrational.}}$ **[1M]**

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Q10. CASE STUDY 03: PRECISION ATHLETICS CIRCULAR RUNNING TRACK [4 Marks]

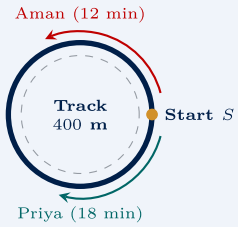
At an inter-state sports complex, an Olympic standard circular synthetic track has a perimeter of 400 m. Two athletes, Priya and Aman, start simultaneously from the same starting marker S and run in the same direction at uniform speeds. Priya takes 18 minutes to complete one full lap, while Aman takes 12 minutes to complete one full lap.

(i) After how many minutes will Priya and Aman meet again at the starting point S? [2]

(ii) How many complete laps will Priya and Aman have run respectively by the time they meet at S? [1]

(iii) If a third athlete Rohan joins them taking 24 minutes per lap, after how many minutes will all three meet at S? [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Circular Motion Condition: Both athletes begin together at S and return to S after exact integer multiples of their lap times (18 min and 12 min), so their earliest reunion is $\text{LCM}(18, 12) = 36$ min.

We know that simultaneous return to the start occurs at the LCM of individual lap times:

Formula Used: Time to reunite at start $= \text{LCM}(T_1, T_2, \dots)$

(i) First meeting of Priya and Aman at S: Prime factorising 18 and 12:

$$18 = 2^1 \times 3^2, \qquad 12 = 2^2 \times 3^1 \implies \text{LCM}(18, 12) = 2^2 \times 3^2 = 4 \times 9 = \boxed{36 \text{ minutes}}. \qquad [2M]$$

(ii) Laps completed by each: Dividing total elapsed time (36 min) by each lap time:

$$\text{Priya: } \frac{36}{18} = \boxed{2 \text{ laps}}, \qquad \text{Aman: } \frac{36}{12} = \boxed{3 \text{ laps}}. \qquad [1M]$$

(iii) With Rohan (24 min): Factorising $24 = 2^3 \times 3$:

$$\text{LCM}(18, 12, 24) = 2^3 \times 3^2 = 8 \times 9 = \boxed{72 \text{ minutes}}. \qquad [1M]$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★